

VIRGINIA KEY, FL, USA

WMO: 997355

Lat: 25.7315N

Lon: 80.1619W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.9	(c) 12.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.4	(k) 22.4	(l) 9.4	(m) 22.5	(n) 3.4	(o) 340	(p) 0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 2.9	(c) 30.8	(d) N/A	(e) 30.4	(f) N/A	(g) 30.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.9	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.3	(b) 8.5	(c) 7.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	20.5	21.8	21.9	24.1	25.7	27.6	28.2	28.6	27.9	26.5	23.5	22.1		
	DBStd	3.56	3.68	3.06	2.70	1.79	1.24	1.16	1.08	0.90	1.03	1.76	2.56	3.34		
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	1		
	HDD18.3	56	24	11	7	0	0	0	0	0	0	0	2	12		
	CDD10.0	5432	326	331	368	424	487	527	563	575	537	512	405	377		
	CDD18.3	2444	91	108	117	174	228	277	305	317	287	253	157	129		
	CDH23.3	22596	194	329	347	993	1957	3237	3843	4117	3462	2616	939	562		
	CDH26.7	6021	0	1	7	42	157	959	1434	1677	1159	533	51	2		
(14) Wind	WSAvg	4.0	3.9	4.0	4.3	4.5	4.3	3.8	3.5	3.5	3.5	4.1	4.0	4.1		
(15) Precipitation	PrecAvg	1512	49	57	59	78	129	231	170	207	235	171	70	57		
	PrecMax	1861	180	139	159	157	285	421	271	321	384	370	232	276		
	PrecMin	1266	8	3	5	11	5	79	92	76	135	41	9	10		
	PrecStd	178	40	38	37	41	69	103	53	61	73	101	60	59		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.9	26.4	27.7	29.5	29.4	31.2	31.1	31.1	31.8	30.3	28.6	26.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	25.5	25.8	26.3	27.6	28.3	30.3	30.5	30.7	30.3	29.5	28.0	26.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	25.1	25.5	25.6	26.9	27.9	29.9	30.2	30.4	30.0	29.0	27.5	26.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	24.6	25.1	25.1	26.5	27.5	29.4	29.9	30.2	29.6	28.6	26.7	25.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	3.8	3.9	3.0	2.7	2.8	2.9	2.9	3.1	3.0	4.0	3.8		
		MCDBR	3.1	3.1	4.0	3.8	3.2	3.2	3.1	3.1	3.4	3.2	3.2	2.6		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.365	0.375	0.387	0.398	0.425	0.457	0.462	0.453	0.430	0.415	0.381	0.374		
	taud		2.527	2.511	2.465	2.418	2.387	2.357	2.338	2.389	2.449	2.460	2.503	2.520		
	Ebn at Noon		889	905	910	902	871	837	833	842	856	853	864	863		
	Edh at Noon		93	101	111	119	122	125	128	121	111	105	94	91		
(44) All-Sky Solar Radiation	RadAvg		3.71	4.51	5.52	6.25	6.29	5.75	5.92	5.69	5.00	4.58	3.93	3.45		
	RadStd		0.25	0.27	0.26	0.29	0.51	0.50	0.26	0.30	0.25	0.33	0.18	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.39	+0.88	N/A	N/A	N/A	N/A	N/A	-20	+135	+114

Nomenclature: See separate page